



DAY 60 · PYTHON LEARNING ROADMAP

Pushing to GitHub & Remote Repositories

A complete, saved record of today's lesson — account setup, commands, and full workflow

Prepared for **Shoaib** (github.com/Muhammad-shoaib-hub)

Day 60: Pushing to GitHub & Remote Repositories

Welcome back, Shoaib! Today is Day 60: Pushing to GitHub & Remote Repositories.

Today is an exciting milestone! Up until now, your Git repositories have lived only on your local computer. Today, you are going to connect your local folder to GitHub—the cloud platform where developers back up their projects, showcase their portfolios, and collaborate with teams.

1. What is a Remote Repository?

- **Local Repository:** The Git project on your laptop or computer.
- **Remote Repository (Remote Repo):** A copy of your project stored on GitHub's cloud servers.
- **Pushing:** Sending your local commits up to GitHub.
- **Pulling:** Downloading changes from GitHub down to your local machine.

2. Essential Commands for Day 60

Action	Command
Add Remote Link	<code>git remote add origin <github-repo-url></code>
Check Remote Link	<code>git remote -v</code>
Push Code to GitHub	<code>git push -u origin main</code>
Push Future Updates	<code>git push</code>

3. Today's Step-by-Step Practical Exercise 🛠️

Let's publish your python learning project to GitHub!

Step 1: Create a New Repository on GitHub

1. Go to GitHub.com and log in.
2. In the top-right corner, click the + icon and select New repository.
3. Set your Repository name to: `python-learning`
4. Choose Public (so anyone or potential employers can view your code).
5. Important: Leave "Add a README file", ".gitignore", and "Choose a license" UNCHECKED (since we already have a local project).
6. Click the green Create repository button.

Step 2: Connect Your Local Folder to GitHub

After creating the repository, GitHub will show you a page with some commands under "...or push an existing repository from the command line".

Open your VS Code terminal in your python learning folder and run these commands:

```
# 1. Ensure your default branch is named main
git branch -M main

# 2. Connect your local folder to your online GitHub repository
# (Replace YOUR_USERNAME with your real GitHub username)
git remote add origin https://github.com/YOUR_USERNAME/python-learning.git

# 3. Push your code to GitHub for the first time
git push -u origin main
```

Give this a try! If GitHub asks you to log in or authorize in the browser during git push, just click allow.

The Conversation, Step by Step

its first time i open it ... so i need to create account on it right?

Yes, exactly! Since this is your first time visiting GitHub, you need to create your free account first.

Step-by-Step: Creating Your GitHub Account

1. Click the Sign up button in the top-right corner (or click the green "Sign up for GitHub" button in the middle of the screen).

2. Enter your email address.
3. Create a strong password.
4. Choose a username (Pick something professional! Employers will see this username, e.g., shoaib-dev or shoaib-coder).
5. Type y or n if it asks about receiving updates, then solve the quick verification puzzle.
6. Check your email for a 6-digit verification code sent by GitHub and enter it on the screen.
7. Once your account is created and you reach the dashboard, let me know so we can link your local project!

i give this name (shoaib-glitch)

shoaib-glitch sounds like a awesome developer username! 😊

Now that your account is set up, let's create your first online repository and link your python learning folder to it:

Step 1: Create the Repository on GitHub

1. Once you are logged in, look at the top-right corner of GitHub and click the + icon, then click New repository (or click the green "Create repository" button on your homepage).
2. For Repository name, type: `python-learning`
3. Make sure Public is selected.
4. IMPORTANT: Do NOT check "Add a README file", ".gitignore", or "Choose a license" (leave them empty/unchecked since we already have a folder on your computer).
5. Click the green Create repository button at the bottom.

Step 2: Push Your Code from VS Code

After clicking create, GitHub will show you a page with commands. Open your terminal in VS Code (inside your python learning folder) and run these 3 commands one by one:

```
git branch -M main
git remote add origin https://github.com/shoaib-glitch/python-learning.git
git push -u origin main
```

(If a pop-up appears asking you to authorize or log into GitHub from VS Code, just click Authorize / Sign In).

it shows this page

Awesome! Your online repository python-learning is ready on GitHub.

Notice in your screenshot that your GitHub URL uses **Muhammad-shoaib-hub** as your username!

Now, let's link your VS Code project to this exact repository.

Run These 3 Commands in VS Code

```
git branch -M main
git remote add origin https://github.com/Muhammad-shoaib-hub/python-learning.git
git push -u origin main
```

(If a browser pop-up asks to sign in or authorize VS Code to access GitHub, click Authorize or Sign In).

it give me this (sign-in popup)

That pop-up is completely normal! That is Git asking for permission to link your computer to your GitHub account securely.

What to Click Now

1. Click the big pinkish-red button: "Sign in with your browser".
2. It will open a tab in your Chrome browser asking you to Authorize Git / VS Code.
3. Click Authorize github (or Confirm).

Once you authorize it in the browser, look back at your VS Code terminal! It will finish uploading your code and print out messages showing that all your files were pushed to GitHub!

this come on the screen (Git Credential Manager)

Perfect! That is the official Git Credential Manager asking you to sign in so it can securely link your computer to GitHub.

What to Do Next

1. Enter your GitHub Username or email address (e.g., Muhammad-shoaib-hub or the email you registered with).
2. Enter your Password.
3. Click the green Sign in button.
4. (If a green button appears on the next screen asking to Authorize Git Credential Manager, click it!)

it give me this (password blocked)

No worries at all, Shoaib! GitHub doesn't allow direct password authentication for Git commands anymore for security reasons.

Here is the easiest way to fix this in under a minute using Sign in with a code:

Step 1: Switch Sign-in Method

1. Close that browser tab or go back to the small black pop-up box that popped up earlier in VS Code (titled "Connect to GitHub").
2. Instead of clicking "Sign in with your browser", click Sign in with a code.

Step 2: Authorize in Browser

1. VS Code will show you an 8-character code (and automatically open a browser page at github.com/login/device).
2. Paste or type that code into the browser box.
3. Click Continue / Authorize.

Once you enter that code, your VS Code terminal will instantly connect to GitHub and finish pushing your code!

it give me this (device code shown)

Perfect! That is the device code you need.

Here is what to do right now:

1. Click on the link at the bottom of that window: <https://github.com/login/device> (or open it in your browser).
2. Type or paste this exact code into the box: `6408-622F`
3. Click Continue and then click Authorize github.

Once you authorize it in the browser, look back at your VS Code terminal! It will automatically complete the push.

it again come to here (wrong browser)

Notice in the top bar of your browser window that you are in Microsoft Edge, but earlier you opened GitHub in Google Chrome!

Since Edge isn't logged into your GitHub account yet, GitHub is asking you to sign in first.

Easy Fix: Use Chrome

1. Open your Google Chrome browser (where you were already logged into GitHub).
2. Open a new tab and go to: <https://github.com/login/device>
3. Type in the code: 6408-622F (or if the code expired, click "Sign in with a code" again in VS Code to get a fresh code).
4. Click Continue -> Authorize github.

Alternatively, if you want to use Edge, you can just click "Continue with Google" on that screen (if you created your account with your Gmail), or type in your username (Muhammad-shoaib-hub) and your GitHub password carefully.

Doing it in Chrome is much faster since your session is already active there!

now what i do?

You are right at the final step! 🎉

What to Click Now

Click the green button at the bottom: Authorize git-ecosystem.

What Happens Next?

1. The browser will display a message like "Success" or "Device connected".
2. Switch back to your VS Code window.
3. Look at your VS Code terminal! You will see Git uploading your files. It will look similar to this:

```
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Writing objects: 100% (15/15), done.
To https://github.com/Muhammad-shoaib-hub/python-learning.git
* [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Go refresh your GitHub page (github.com/Muhammad-shoaib-hub/python-learning) in your browser—all your Python files and folders will now be live online!

Authorize git-ecosystem — but it can't click

If the green Authorize git-ecosystem button isn't clicking or responding, don't worry—this is usually caused by browser extension

interference (like an ad-blocker) or a temporary focus glitch.

Here are 3 quick ways to get past this step:

Option 1: Scroll down or zoom out in the browser

1. Click anywhere on the browser page.
2. Scroll down to make sure the entire button is visible.
3. Try zooming out slightly by pressing Ctrl + - (or holding Ctrl and scrolling your mouse wheel down), then click the green Authorize git-ecosystem button again.

Option 2: Try in Chrome (Incognito / Private Mode)

1. Open a new Incognito / Private Window in Google Chrome (Ctrl + Shift + N).
2. Go to <https://github.com/login> and sign into your Muhammad-shoaib-hub account.
3. In that same incognito window, open a new tab and go to <https://github.com/login/device>.
4. In VS Code, click Sign in with a code again to get a fresh code.
5. Paste the fresh code into the incognito browser tab and click Authorize.

Option 3: Authorize directly inside VS Code

If the external browser keeps getting stuck:

1. Close the browser tab.
2. Go back to VS Code.
3. Look at the bottom-left corner of VS Code for the Accounts icon (the small user circle icon above the gear icon).
4. Click it and select "Sign in with GitHub".
5. Follow the quick prompt directly inside VS Code!

yes sir it show this ("Something went wrong")

Sorry, something went wrong. Please try your request again.

it show this ("Congratulations, you're all set!")

Awesome job, Shoaib! 🎉 "Congratulations, you're all set!" means your device is now officially connected to your GitHub account!

What to Do Right Now

1. Switch back to VS Code: Click on your VS Code icon in the taskbar.
2. Check your Terminal: You should see Git finishing up the upload process automatically.
3. Verify on GitHub: Go to your browser tab with your repository link (<https://github.com/Muhammad-shoaib-hub/python-learning>) and press F5 (or refresh the page).

You should now see all your local files and folders uploaded live on GitHub!

yes sir it shows this (final terminal output)

```
PS C:\Users\shoaib\Desktop\python learning> git push -u origin main
Enumerating objects: 92, done.
Counting objects: 100% (92/92), done.
Delta compression using up to 4 threads
Compressing objects: 100% (88/88), done.
Writing objects: 100% (92/92), 116.07 KiB | 1.71 MiB/s, done.
Total 92 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (6/6), done.
To https://github.com/Muhammad-shoaib-hub/python-learning.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

CONGRATULATIONS, SHOAIB! 🎉 Day 60 is officially a success!

Look at those final lines:

```
* [new branch] main -> main
branch 'main' set up to track 'origin/main'
```

That means all 92 objects (all your code, scripts, and practice files from your python learning folder) have been safely uploaded to GitHub's servers!

Take a Look at Your Work Live 🌐

Go back to your browser, open your repository link, and refresh the page (or click the link below):

🔗 <https://github.com/Muhammad-shoaib-hub/python-learning>

You will see all your files listed online, complete with your commit history!

Quick Recap of the Workflow (For Future Days)

From now on, whenever you make changes or add new Python files, your daily workflow will be super simple:

```
# 1. Stage your changes
git add .

# 2. Commit your changes
git commit -m "Add Day 61 exercise"

# 3. Push to GitHub
git push
```

(Notice you won't need to sign in again or use -u origin main—Git now remembers your connection!)

You've built your first public remote repository! How does it feel seeing your code live on GitHub?

i close the all browser ,, tell me how i will check my git-hub account now please

Checking your GitHub account now is super easy!

Step-by-Step Instructions

1. Open your browser (Google Chrome or Microsoft Edge).
2. Go to: github.com/Muhammad-shoaib-hub
3. You will see your profile page! Under Popular repositories or the Repositories tab at the top, click on python-learning.
4. Alternatively, you can go straight to your project anytime using this direct link:

🔗 <https://github.com/Muhammad-shoaib-hub/python-learning>

Tip: You can bookmark that URL in your browser (Ctrl + D) so you can open it with one click whenever you want to check your progress!

it shows this now (repository live)

That looks fantastic! Everything is live on your profile right where it belongs:

- Your folders & files: backend, day51, and WhatsApp Image... are all listed in your python-learning repository.
- Commits: You can see your recent commit message ("i just added day59.py on shoaib-branch") logged right next to the files.

Everything is set up properly and linked between your local machine and your remote GitHub account!

★ Complete GitHub Setup & Workflow Guide

Here is a complete, step-by-step summary guide of everything we covered today. You can copy and paste this into a file or note (like a README.md or a Word document) to save for your records!

Part 1: How to Create a GitHub Account

1. **Visit GitHub:** Go to github.com.
2. **Sign Up:** Click the Sign up button in the top right corner.
3. **Enter Details:**
 - Enter your Email address.
 - Create a strong Password.
 - Pick a Username (e.g., Muhammad-shoaib-hub).
4. **Verify & Create:** Complete the quick puzzle to verify you are human, click Create account, and enter the launch code sent to your email.

Part 2: What We Did Today (Day 60 Milestones)

- **Created a Remote Repository:** Created a public online repository named python-learning on your GitHub account (<https://github.com/Muhammad-shoaib-hub/python-learning>).
- **Connected Local Project to Remote Repository:** Linked your local python learning desktop folder to your online GitHub repository.
- **Authenticated VS Code:** Authenticated Git using the Git Credential Manager via Device Code Authentication (Sign in with a code).

- **Pushed Code Online:** Successfully pushed 92 objects (files and folders) from your computer to the main branch on GitHub.

Part 3: Today's Git Commands Reference

Here are the exact commands executed today to set up and push your code:

1. Set the primary branch to main

```
git branch -M main
```

2. Link your local project to GitHub

```
git remote add origin https://github.com/Muhammad-shoaib-hub/python-learning.git
```

3. Push local files to GitHub for the first time

```
git push -u origin main
```

Part 4: Daily Workflow for Future Learning Days

From now on, whenever you write new code or make changes to your files, just run these 3 simple commands in your VS Code terminal:

```
# 1. Stage all your new/updated files
git add .

# 2. Save a snapshot of your changes with a message
git commit -m "Add Day 61 Python practice"

# 3. Upload your changes to GitHub
git push
```

🔗 Your Live Repository Link: <https://github.com/Muhammad-shoaib-hub/python-learning>